Introduction to (Seasonal) Autoregressive Integrated Moving Average Model: Implementation in R

Math 324: Statistics
Final Project

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Abstract

This report presents the examination of the ARIMA Model (Autoregressive Integrated Moving Average Model) and SARIMA Model (Seasonal Autoregressive Integrated Moving Average Model) in depth, concentrating on their implementation in R on two distinct empirical datasets: the US GDP dataset, and the monthly mean carbon dioxide dataset. We begin with the explanatory data analysis and progress to the model selection and estimation. ARIMA(2,0,2) is selected for the short-term forecast of the US GDP dataset, and ARIMA(1,1,1) $\times(0,1,1)_{12}$ is selected for the short-term forecast of the monthly mean carbon dioxide dataset. The limitations of our study will be discussed based on the forecast results, including the impact of outlier and non-periodic trends. This report also provides suggestions for future studies.

1 Introduction

A discrete time series is one in which observations are recorded at equally spaced intervals. In mathematical terms, a discrete time series is defined on a discrete index set, such as the set of integers or natural numbers. Time series analysis can be explained as an analysis of the correlation between the adjacent data points of a certain sample. Finding the most accurate time series forecasting model is meaningful and comprehensive. It has a significant effect in various fields, including financial time series Gross Domestic Product (GDP) and the measurement of atmospheric carbon dioxide levels. GDP contains diverse economic information, and forecasting provides guidance for economic policy changes and prepares for solving global problems such as inflation. Forecasting of carbon dioxide level is also meaningful as increasing the amount of carbon dioxide in the atmosphere increases the greenhouse effect, and further causes global warning. Measuring amount of carbon dioxide and forecasting its growth rate can help people take action to limit climate change.

Optimizing the best forecast model is challenging. There are many forecasting models for time series data, for example, grey prediction, ARIMA (Auto-Regressive Integrated Moving Average), and SARIMA (Seasonal - ARIMA) model, also some machine learning models like random forest (RF) [Li and Zhang, 2023]. Different methods are being used for the data with different features. ARIMA and SARIMA models combined with the Box-Jenkins methodology are being widely used for time series forecasting.

There are many studies that have been carried out that use the ARIMA model to understand GDP for various countries. There is a previous study suggesting that ARIMA(1,2,1) is the appropriate model for Egyptian GDP [Abonazel and Abd-Elftah, 2019], and ARIMA(2,2,2) is the best ARIMA model for Kenyan [Wabomba, 2016]. The GDP of Bangladesh has been fitted into ARIMA(1,2,1) [Miah, 2019]. Numerous projects have been devoted to different forecasting models on the carbon dioxide time series data, and some have suggested that machine learning techniques have a better prediction result compared with traditional methods like the SARIMA model [Li and Zhang, 2023]. However, this study has demonstrated that SARIMA is a suitable model for forecasting atmospheric carbon dioxide levels [Gogeri et al., 2023].

In this project, we want to find the best ARIMA model for the US GDP dataset, and the SARIMA model for the carbon dioxide dataset, make a forecast based on those data and draw a conclusion.

2 ARIMA Models

2.1 Definiton

The Autoregressive Integrated Moving Average (ARIMA) model is a widely used forecasting technique for time series data. It is built upon the Autoregressive Moving Average (ARMA) model, which is designed for stationary time series. ARIMA generalizes ARMA by incorporating an initial differencing step (the "I") to transform a non-stationary time series into a stationary one, making it applicable to a broader range of data.

- **Weakly Stationary:** A weakly stationary time series refers to a time series that has a constant mean and variance that doesn't vary over time. Also, the auto-covariance function of a stationary time series depends only on the absolute value of the difference between two time points.
- **Trend:** An increase (upward trend) or a decrease (downward trend) is usually referred to when the plot shows a long-term increase or decrease.
- **Cyclic:** The fluctuation is not periodic.
- **The autocorrelation function (ACF):** ACF is used to indicate the linear predictability of x_t, only based on x_{t-k}, it implies the correlation between observations at different lags

$$ACF(k) = \frac{\sum_{t=k+1}^{n}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{n}(y_t - \bar{y})^2}$$

- **The partial autocorrelation function (PACF):** PACF implies the correlation between the time series and the lag values of itself.

$$PACF(k) = \frac{\gamma(k|k)}{\sqrt{(\gamma|k) \cdot \gamma(0|k)}}$$

2.1.1 Autoregressive (AR)

Autoregressive models imply that the current value of the times series x_t can be explained as a linear function of p past values, $x_{t1}, x_{t2}, ..., x_{tp}$. We need p number of steps into the past to forecast the current value. The p is the order of the AR model.

The expression of AR(p) model is:

$$x_t = \Phi_1 x_{t-1} + \Phi_2 x_{t-2} + ... + \Phi_p x_{t-p} + w_t$$

where x_t is a stationary series data points, w_t is a white noise and $\Phi_1, \Phi_2, ..., \Phi_p$ are constants.

2.1.2 Integrated Models for Nonstationary Data (I)

Differencing transforms nonstationary time series data by replacing values with the changes between consecutive observations. For first-order differencing we have:

$$\Delta x_t = x_t - x_{t-1}$$

For higher-order differencing, the process is applied repeatedly to get the following form:

$$\Delta^d x_t = (1 - B)^d x_t$$

where B is defined as $Bx_t = x_{t-1}$. The resulting differenced series is stationary and can be modeled as:

$$\Delta^d x_t = \epsilon_t$$

where x_t is non stationary time series, ϵ_t is the resulting stationary time series and Δ^d is the differencing operator of order d. This equation indicates that the d-th difference of the time series x_t follows a stationary process.

2.1.3 Moving Average (MA)

x_t on the left-hand side of the equation are assumed to be combined linearly according to the AR model, while the right side of the equation is white noise which are combined linearly to form the observed data.

The expression of $MA(q)$ is:

$$x_t = \Theta_1 w_{t-1} + \Theta_2 w_{t-2} + ... + \Theta_q w_{t-q} + w_t$$

where w_t is a white noise and $\Theta_1, \Theta_2, ..\Theta_q$ are parameters

2.2 Parameters

p: The order of the Autoregressive model is the number of past values (lags observations) we need to forecast the current value of the time series.

d: The order of differencing, the number of differencing we need to make the time series stationary.

q: The order of the Moving Average; is the number of lags forecast errors we need to forecast the current value of the time series.

3 SARIMA Models

3.1 Definition

SARIMA model is an extension of the non-seasonal ARIMA model, to deal with the data with seasonal fluctuation. SARIMA model has the seasonal autoregressive term and the seasonal moving average term that could identify the seasonal lags.

3.1.1 Seasonal Autoregressive (SAR)

Seasonal Autoregressive implies how the current value of the time series can be explained by the past values of the time series data. We need P number of past values at seasonal lags to forecast the current value.

The expression of the Seasonal Autoregressive model (P) is:

$$x_t = \Phi_t x_{t-s} + \Phi_2 x_{t-2s} + ... + \Phi_p x_{t-ps} + w_t$$

where x_t is the time series, w_t is the white noise and $\Phi_1, \Phi_2, ..., \Phi_p$ are constants.

3.1.2 Seasonal Integrated Models for Seasonal Data (SI)

The number of seasonal differences we need to make the time series data non-seasonal. For first-order differencing we have:

$$\Delta_s x_t = x_t - x_{t-s}$$

For higher-order differencing, the process is applied repeatedly to get the following form:

$$\Delta_s^D x_t = (1 - B^s) x_t$$

where B is defined as $Bx_t = x_{t-1}$. The resulting differenced series is stationary and can be modeled as:

$$\Delta^d \Delta_s^D x_t = \epsilon_t$$

where x_t is a non stationary time series, ϵ is the resulting stationary time series and Δ^D is the seasonal differencing of order D, Δ_s^t is the regular differencing operator of order d.

3.1.3 Seasonal Moving Average (SMA)

Seasonal Moving Average implies how the current value of the time series can be explained by the forecast errors at the seasonal lags.

The expression of the Seasonal Moving Average model(Q) is:

$$x_t = \Theta_1 w_{t-s} + \Theta_2 w_{t-2s} + ... + \Theta_q w_{t-qs} + w_t$$

where w_t is a white noise and $\Theta_1, \Theta_2, ..\Theta_q$ are parameters.

2.2 Parameters

P: The order of the Seasonal Autoregressive model is the number of past values (lags observations) at seasonal lags we need to forecast the current value of the time series.

D: The order of Seasonal differencing, the number of differencing we need to make the time series non-seasonal.

Q: The order of the Seasonal Moving Average is the number of lags forecast errors we need to forecast the current value of the time series.

s: The length of the seasonal period and it is also the number of observations for one year.

4 Data Analytic Strategies

4.1 Identify Time Series Pattern

After constructing the plot of data, it is important to unveil the patterns of the time series so that we can further apply the necessary transformations to the data.

To detect the stationary of the data, we could look at the Auto-correlation Function (ACF plots). The values of lags tend to decay to zero more quickly for stationary data. The values of logs of non-stationary data will decay to zero, and have many lags with values close to one.

Several graphical techniques can be used to detect the seasonality of the time series, for example, running sequence plots. This can show both the seasonal difference between different group patterns, and also the within-group patterns. We can also look at the ACF; at specific lags, the existence of some regular peaks in the ACF plot indicates the seasonality of the time series.

4.2 Data Transformation

The most important step involved in data transformation is eleminating the trend and the seasonality of time series.

We will apply the logarithmic transformation to stabilize the variance to eliminate the upward or down- ward trend of the variance and stabilize the variance.

To eliminate the trend, we will take the differencing by computing the differences between the two consecutive time points. If there needs additional difference, we could apply second order difference, but it's important to avoid over-difference. Differencing will stabilize the mean of the time series and further transfer the non-stationary data to stationary data.

To eliminate seasonality, we will take the seasonal differencing by computing the differences between the current time observation and the previous time observation at the same season. This will eliminate the seasonal component of the time series, and further transfer the data to stationary.

4.3 Parameters Identification and Estimation

After and the transform the data, we will start to identify the order of the model and estimate the parameters.

4.3.1 ARIMA Model

p is the order of the Autoregressive model, d is the order of differencing, and q is the order of the Moving Average model. To estimate the value of p, d and q, we will look at ACF and PACF plots of certain time series data.

	AR(p)	MA(q)	ARMA(p,q)
ACF	Tails off	Cuts off after lag q	Tails off
PACF	Cuts off after lag p	Tails off	Tails off

Table 1: Behavior of the ACF and PACF for ARMA Models

Recall that **cuts off** describes a sudden drop of the autocorrelation to zero after a given lag, indicating no further significant dependence. In contrast, **tails off** describes a slow, gradual decline in correlation as the lag increases.

This theoretical framework will be applied more concretely in the subsequent ARIMA analysis of US GDP data to determine the optimal model order.

We will decide and estimate the value of the parameters according to Table 1 above, and then we will test the significance of the parameters.

To test the significance level of the coefficients that we estimate, we will compute the t-statistics and p-value. The null hypothesis is that there is no significant difference between the value of the coefficient and zero. The p-value smaller than 0.5 implies that this coefficient that we estimate is significant. A p-value that is greater or equal to 0.5 implies that this coefficient is not significant, there will need some adjustment to the current model.

4.3.2 SARIMA Model

p, q and d are the parameters for the non-seasonal part, and P is the order of the seasonal Autoregressive model, D is the order of seasonal differencing, and Q is the order of the seasonal Moving Average model. To estimate the value of p, q, d, P, D and Q, we will look at ACF and PACF plots of certain time series data.

	AR(P)	MA(Q)	SARMA(P,Q)
ACF*	Tails off at lags ks	Cuts off after lag Qs	Tails off
PACF*	Cuts off after lag Ps	Tails off	Tails off at lags ks

Table 2: Behavior of the ACF and PACF for SARMA Models

We will decide and estimate the value of P, D, and Q according to Table 2 above, to decide the p,d, and q, we will inspect the ACF and PACF plots at the lower lags. To test the significance of these parameters, we will use T-statistics and p-value as how we test the significance level for the parameters of the ARIMA model.

4.4 Model Diagnostics

Residuals are the difference between the observed value and the fitted value. Good forecasting needs the residual to be uncorrelated, to have a zero mean, constant variance variance and to be normally distributed.

4.4.1 Standard Residuals

Standard residuals plots can be used to check if the residuals have a constant variance with a zero mean. The existence of a rectangular scatter around a zero horizontal y-axis will support the model.

4.4.2 ACF of Residuals

This plot can be used to test if there is a correlation between the residuals. If there are beyond 5% lags that have a value that is larger than the significant level, this might imply that we lost some significant information within the residuals which should be accounted for in the model.

4.4.3 Normal Q-Q Plot of Standard Residuals

Normal Q-Q Plots of Standard Residuals are being used to check the normality of the residuals. The residual points need to coincide with the straight line in the plot. The outliers that do not coincide with the line might imply that the residuals are not roughly normally distributed.

4.4.4 p-value for Ljung-Box Statistics

Ljung-Box statistics are also being used to test the autocorrelation between the residuals. The null hypothesis is that there is no autocorrelation between the residuals. The larger p-value implies the failure to reject the null hypothesis, which indicates residuals are not auto-correlated.

4.5 Model Choice

If we have more than one model that can be considered a good forecasting method after the diagnostics analysis. We will make the model choice based on model selection criteria, AIC (Akaike Information Criterion), AICc (corrected Akaike Information Criterion), and BIC (Bayesian Information Criterion). AIC, AICc, and BIC are used to find the best model by balancing goodness-of-fit with model complexity, ensuring the chosen model explains the data well without being unnecessarily complex. We will choose the model with the smaller AIC, AICs, and BIC values.

AIC measures the goodness of fit for a model. AICc is the AIC version that has been corrected for a small sample size (usually when $\frac{n}{k} < 40$). BIC tends to measure the goodness of fit for simple models due to the smaller penalty term.

- AIC $= -2ln(L) + 2k$
- AICc $= -2ln(L) + 2k + \frac{2k(k+1)}{n-k-1}$
- BIC $= -2ln(L) + kln(n)$

4.6 Forecasting

We will use the appropriate model that we have decided to forecast the future data of the time series under a 95 % confidence interval. The output of an ARIMA model typically includes the point estimate of the forecast value. It is the best-predicted value of the model for the future value of time series at a certain time point.

For the forecast. ARIMA() in R, the function will provide the confidence interval along with the point estimate for each future data point in the time series. The future point estimated value is forecast based on the past data and the parameters of the model that have been fitted into historical time series data points. For each forecasting time point, the function will output the upper bound and lower bound for the confidence interval for the estimated value, which is calculated based on the forecasting error's estimated variance and the given level of confidence. The forecasting usually has a lower accuracy, and the confidence interval has a larger range for the forecasting time point that is further into the future.

5 Case Study 1 ARIMA Model

Gross Domestic Product is the monetary value produced through the production of goods and services by labor and property, which can be used to track the health level of a certain country. This dataset is US GDP that is recorded quarterly from 1947.1.1 to 2025.1.1 and was published by the U.S. Bureau of Economic Analysis [U.S. Bureau of Economic Analysis, 2025]. The data are already being seasonally adjusted at an annual rate. Therefore, there is no seasonality in this dataset. In this section, we performed time series data analysis and ARIMA model forecasting on it.

5.1 Exploration Data Analysis

The top panel of Figure 1 shows the plot of US GDP data. There is a strong upward trend, and the mean is increasing as time increases. There is a large dip in the year 2020, compared with the previous year. This is due to the effect of COVID-19. The Business Cycle Dating Committee of the National Bureau of Economic Research (NBER) has reported that in 2020, COVID-19 will produce a huge contraction in economics. There is a decline of 5.1 % at an annual rate in the first quarter; and 31.2 % in the second quarter.

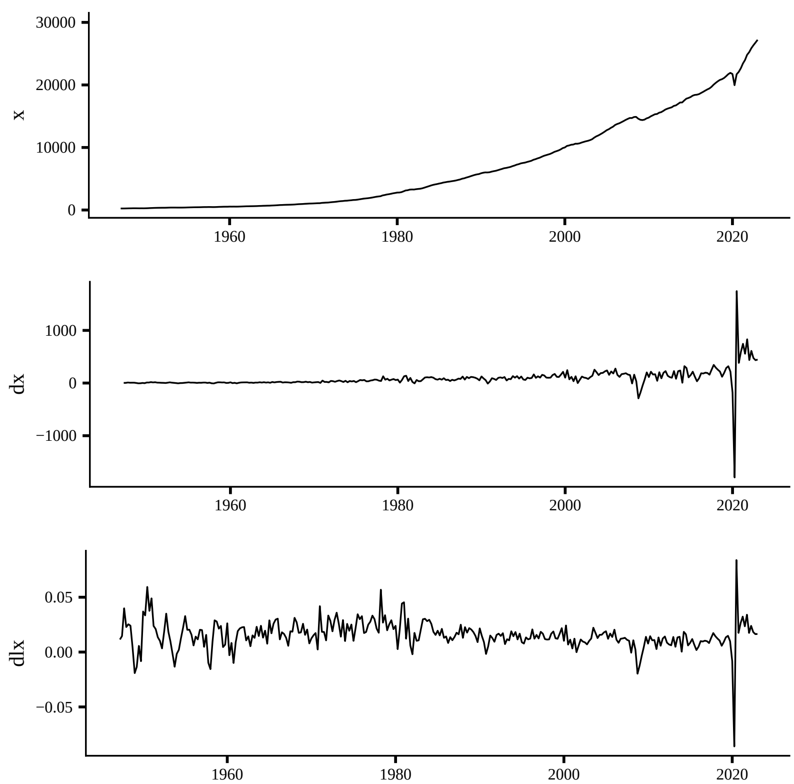


Figure 1: The plot of GDP, diff(GDP), and log(diff(GDP))

The upward trend suggests that the time series data is not stationary. We apply differencing to the original US GDP data to remove the trend, which involves subtracting the current observed value from the previous one. The middle panel of Figure 1 shows the data after differencing, in which the variance is not constant. We apply the logarithmic transformation to the dataset to stabilize the variance. The bottom panel of Figure 1 shows that the mean and variance are approximately constant over time, except there is one outlier at roughly the year 2020 due to the COVID-19 pandemic. Therefore, the following data analysis will be applied to the diff(log(GDP)).

5.2 Identification of the Dependence Orders of the Model

We plot in Figure 2 the ACF and PACF for the diff(log(GDP)) to identify the dependence orders of the model. We set d equal to 0 since we will do the forecasting with the ARIMA model on the differenced data. Both the ACF and PACF exhibit similar patterns, with significant correlations at lags 1 and 2, followed by a sharp cutoff to insignificant values at lag 3. The ACF cuts off after lag 2, suggesting an MA(2) process, while the PACF cuts off after lag 2, suggesting an AR(2) process. Given these complementary patterns, we will fit ARIMA(2,0,0), ARIMA(0,0,2), and ARIMA(2,0,2) for US GDP data to determine the most appropriate model.

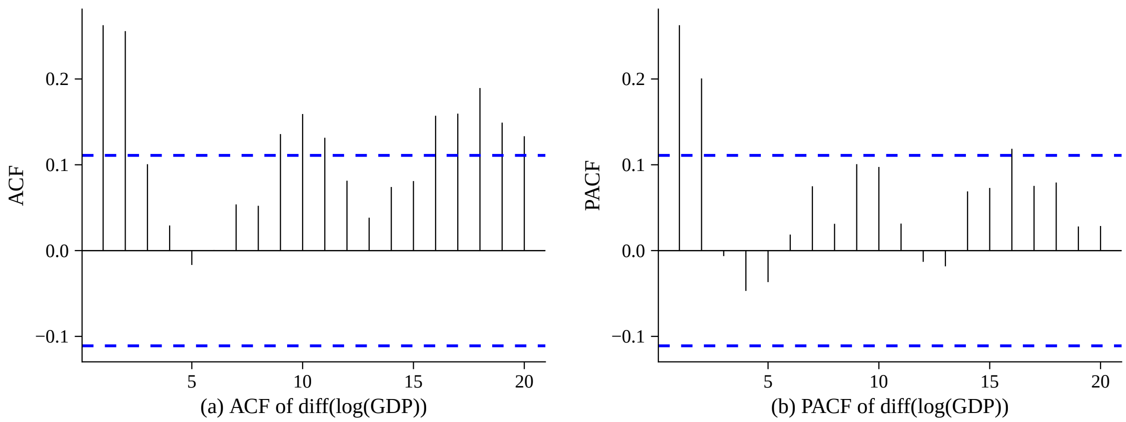


Figure 2: Plots of correlation functions

5.3 Parameter

The parameter estimates for the ARIMA models are presented in Tables 3–5. Each model captures different aspects of the dynamics in the differenced log-GDP series.

5.3.1 ARIMA (2,0,0)

After fitting ARIMA(2,0,0) model, the estimated model is:

$$x_t = 0.0154 + 0.2097x_{t-1} + 0.1999x_{t-2} + w_t$$

with $w_t \overset{\text{iid}}{\sim} (0, \sigma^2)$, $\sigma \approx 0.012058$

Both AR coefficients are positive and statistically significant (p values < 0.05), indicating a moderate and persistent effect of the previous two values on the current observation. The intercept represents the baseline level of the series.

Parameter	Estimate	t-value	p-value
ar1	0.2097	3.79	1.51×10^{-4}
ar2	0.1999	3.61	3.0×10^{-4}
xmean	0.0154	13.35	0

Table 3: Parameter estimate for ARIMA(2,0,0)

5.3.2 ARIMA (0,0,2)

After fitting ARIMA(0,0,2) model, the estimated model is:

$$x_t = 0.0154 + 0.1895w_{t-1} + 0.2234w_{t-2} + w_t$$

with $w_t \overset{\text{iid}}{\sim} (0, \sigma^2)$, $\sigma \approx 0.012112$

Both MA coefficients are positive and statistically significant, indicating a persistent effect of previous forecast errors.

Parameter	Estimate	t-value	p-value
ma1	0.1895	3.37	7.46×10^{-4}
ma2	0.2234	4.23	2.28×10^{-5}
xmean	0.0154	15.90	0

Table 4: Parameter estimate for ARIMA(0,0,2)

5.3.3 ARIMA (2,0,2)

After fitting ARIMA(2,0,2) model, the estimated model is:

$$x_t = 0.0154 + 1.3071x_{t-1} - 0.7345x_{t-2} - 1.1234w_{t-1} + 0.6999w_{t-2} + w_t$$

with $w_t \overset{\text{iid}}{\sim} (0, \sigma^2)$, $\sigma \approx 0.012017$

The AR coefficients suggest a strong autoregressive structure with a damped oscillatory pattern, while the MA coefficients capture alternating corrections from past forecast errors. The intercept remains 0.0154 since we are still working with the same dataset.

Parameter	Estimate	t-value	p-value
ar1	0.1895	3.37	8.49×10^{-12}
ar2	-0.7345	-3.43	6.09×10^{-4}
ma1	-1.1234	-4.80	1.58×10^{-6}
ma2	0.6999	3.37	7.44×10^{-4}
xmean	0.0154	16.74	0

Table 5: Parameter estimate for ARIMA(2,0,2)

5.4 Diagnostics

Figure 3, Figure 4 and Figure 5 show the diagnostics plots for ARIMA (2,0,2), ARIMA (2,0,0) and ARIMA(0,0,2), respectively. There is no obvious pattern in the standards residual plots. There exists a rectangular scatter around a zero horizontal y-axis. There is one obvious outlier in the year 2020. This indicates that the variance of the residuals is roughly constant, and the mean is around zero except for the year 2020. Most of the points coincide or around the straight line in the Q-Q plot except the points at the two ends of the line. This indicates that the residuals are approximately normally distributed.

For the ACF of the residuals, all the lags are less than the 5% significance level. This implies that there is no correlation between the residuals. Most of the p-values for Ljung-Box statistics are above the significant level, which indicates we failed to reject the null hypothesis that there is no correlation between residuals.It provides strong evidence for showing residuals are not auto-correlated.

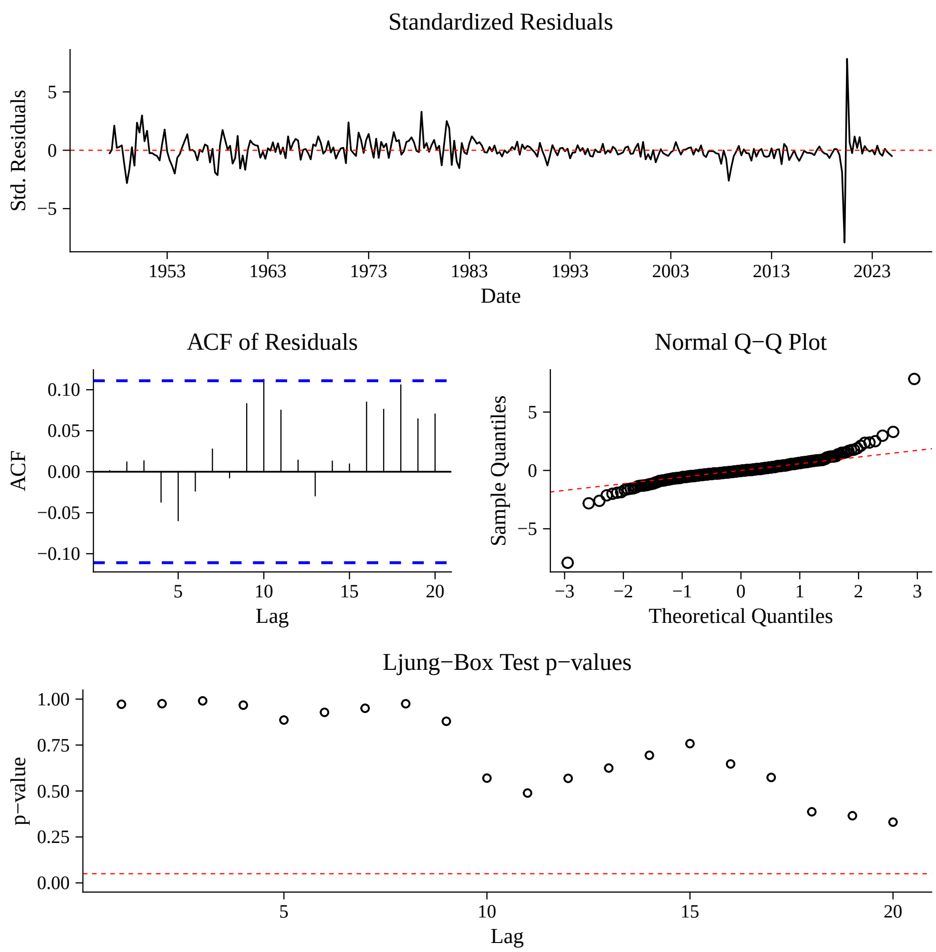


Figure 3: Model diagnostic plots ARIMA for (2,0,0)

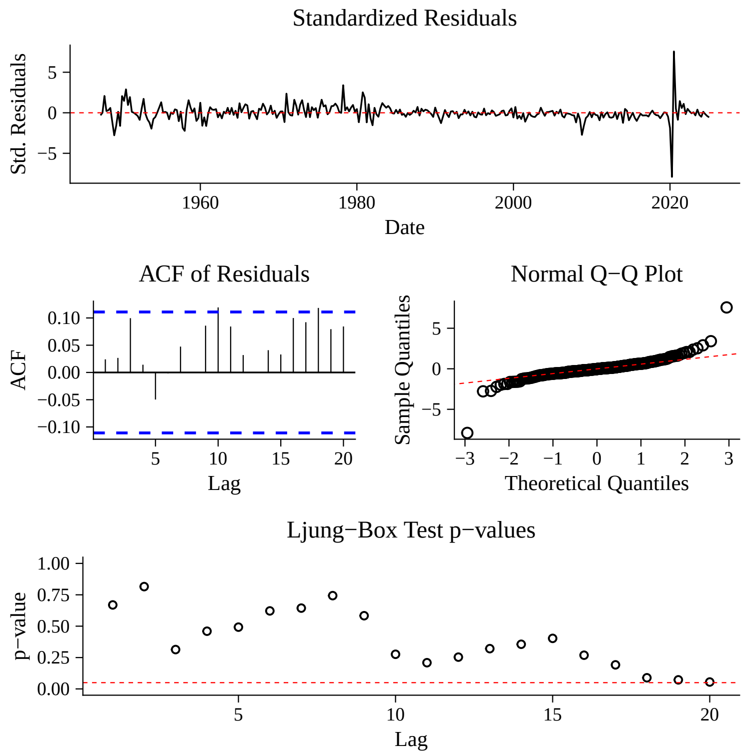


Figure 4: Model diagnostic plots ARIMA for (0,0,2)

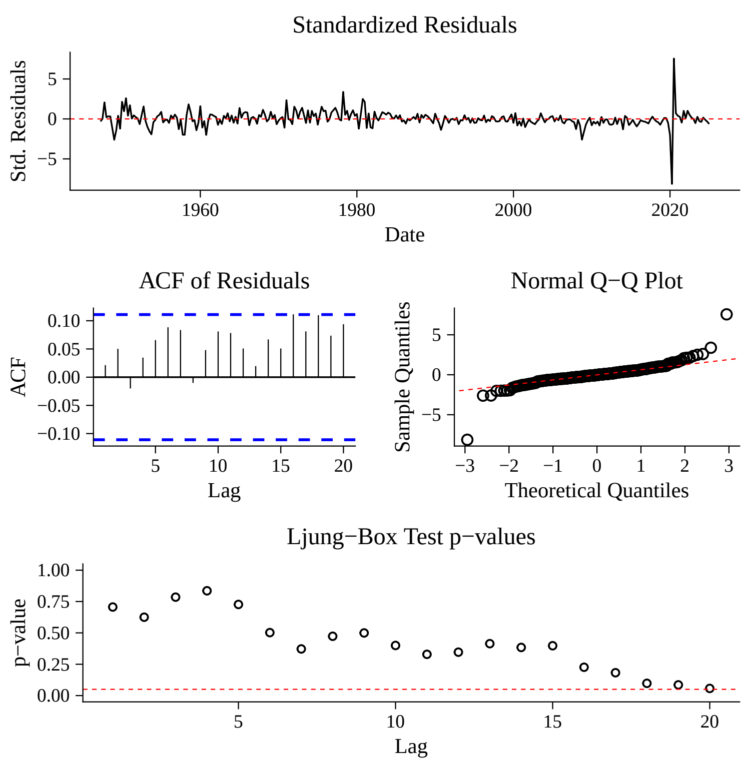


Figure 5: Model diagnostic plots ARIMA for (2,0,2)

5.5 Model Choice

As shown in Table 6, the values of AIC, AICc and BIC of ARIMA(2,0,2) are smaller than the other ARIMA models, so we will use ARIMA(2,0,2) to forecast the future US GDP

Model	AIC	AICc	BIC
ARIMA(2,0,0)	-1861.33	-1861.06	-1838.88
ARIMA(0,0,2)	-1860.43	-1860.3	1845.46
ARIMA(2,0,2)	**-1863.36**	**-1863.23**	**-1848.38**

Table 6: AIC, AICc and BIC of models

5.6 Forecasting

We use ARIMA(2,0,2) to forecast future US GDP. Table 7 show the estimated US GDP value for 9 quarters, with a 95% confidence interval. The confidence level is 95% which measures the degree of uncertainty and the statistical significance of the estimate. It indicates that the US GDP will have a probability of 95% fall into the corresponding confidence interval.

Quarter	Forecast (Billions USD)	Lower 95% CI	Upper 95% CI
2025-Q1	30466.53	29751.76	31198.47
2025-Q2	30904.81	29460.01	32420.47
2025-Q3	31370.07	29176.11	33729.01
2025-Q4	31863.83	28908.56	35121.21
2026-Q1	32378.23	28654.67	36585.66
2026-Q2	32901.73	28402.24	38114.03
2026-Q3	33424.98	28141.19	39700.86
2026-Q4	33944.40	27870.11	41342.57
2027-Q1	34462.34	27593.56	43040.95

Table 7: US GDP Forecast for 2025-2027

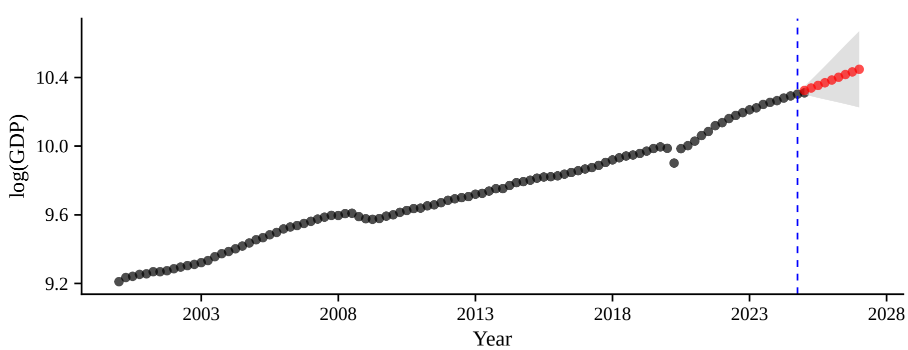


Figure 6: US GDP Forecast 2000-2027

The estimated value of U.S. GDP shows a steady upward trend from 2025 Q1 to 2027 Q1. The 95% confidence interval widens gradually over time, indicating greater uncertainty in long-term projections. This reflects that forecasts further into the future are less precise. Figure 6 illustrates a consistent upward pattern in the log(GDP) plot, suggesting that U.S. GDP is expected to grow gradually and steadily over the forecast period.

6 Case Study 2 SARIMA Model

The time series is the monthly mean atmospheric carbon dioxide concentration (in ppm), which is being measured at Mauna Loa Observatory, Hawaii [Laboratory, 2025]. It is the longest record of direct measurements of carbon dioxide in the atmosphere, from 1958 to 2025. In this section, we performed time series data analysis and SARIMA model forecasting on the monthly mean carbon dioxide dataset.

6.1 Exploration Data Analysis

The plot of raw data in Figure 7 shows that the value of mean carbon dioxide increases sharply over time at a roughly constant rate with increasing mean. The data are not stationary as there is an upward trend. The first-order difference, shown in the middle panel of Figure 7, needs to be applied to eliminate the trend and stabilize the mean.

The periodic fluctuations suggest that there is a seasonality of the monthly mean carbon dioxide data. Therefore, a twelfth-order difference is applied, as shown in the bottom panel of Figure 7, and the transformed data from non-stationary to stationary. The variance of data for each periodic fluctuation is approximately constant, so it is not significant to take the logarithmic transformation.

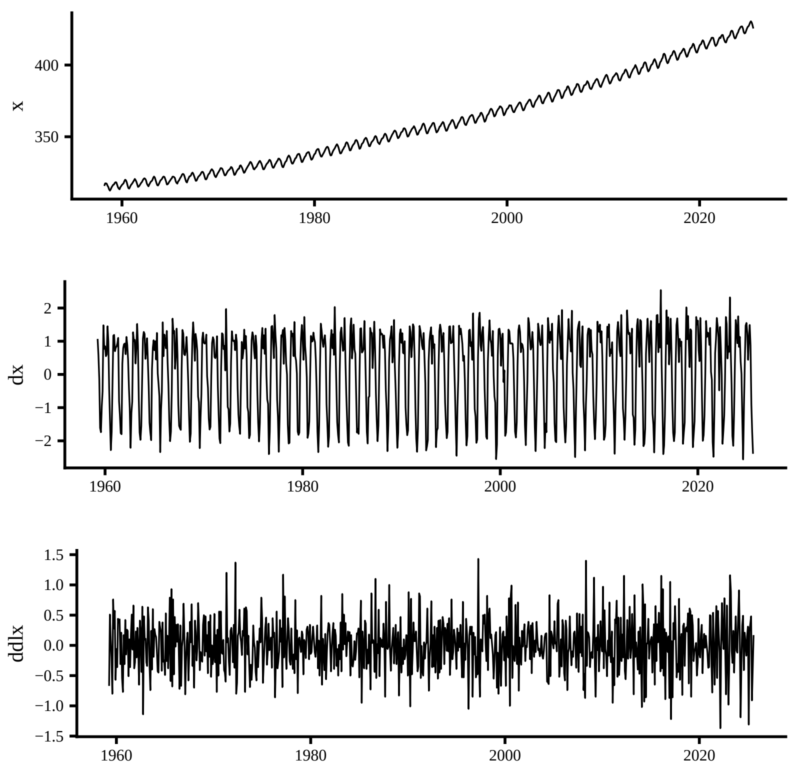


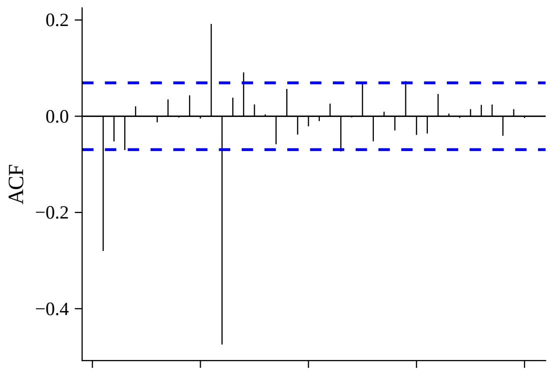
Figure 7: The plot of CO_2, diff(CO_2), and diff(diff(CO_2), lag=12)

6.2 Identification of the Dependence Orders of the Model

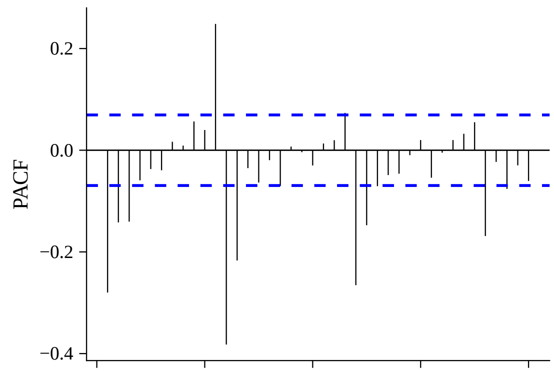
For the seasonal component: at the seasonal lags, ACF cut off at lag 12 then we have a seasonal Q = 1. PACF is tailing off at 12, 24, 36 etc... This implies that P = 0

For the non-seasonal component: focusing at lower lags, both ACF and PACF are tailing off then we have p = q = 1

Therefore, we will try to fit the ARIMA(1,1,1) $\times(0,1,1)_{12}$ model on the transformed data.



(a) ACF of diff(diff(CO2), lag = 12)



(b) PACF of diff(diff(CO2), lag = 12)

Figure 8: Plots of ACF and PACF for transformed CO_2 data

6.3 Parameter Estimation

After fitting the SARIMA(1,1,1) $\times(0,1,1)_{12}$ model, the estimated model is:

$$x_t = 0.2232x_{t-1} - 0.5801x_{t-2} - 0.8682x_{t-12} + w_t$$

σ_w estimated as 0.317 with 793 degree of freedom

Parameter	Estimate	t-value	p-value
ar1	0.2232	2.65	8.14×10^{-3}
ma1	-0.5801	-8.22	2.22×10^{-16}
sma1	-0.8682	-50.06	0

Table 8: Parameter estimate for ARIMA(2,0,0)

6.4 Diagnostics

Figure 9 shows the diagnostic plots for the ARIMA(1,1,1) $\times(0,1,1)_{12}$ model. There are no obvious patterns in the standard residuals plot, there exists a rectangular scatter around a zero horizontal y-axis, suggesting that the variance of the residuals is roughly constant with a zero mean. Most of the points coincide or around the straight line in the Q-Q plot except the points at the two tails of the line. This indicates that the residuals are approximately normally distributed.

For the ACF of the residuals, all algs are lass than the 5% significant level. This implies that there is no correlation between the residuals. All p-values for Ljung-Box statistics are larger than the significant value, which indicates the we failed to reject the null hypothesis that there this no correlation within residuals, showing that the residuals are not auto-correlated.

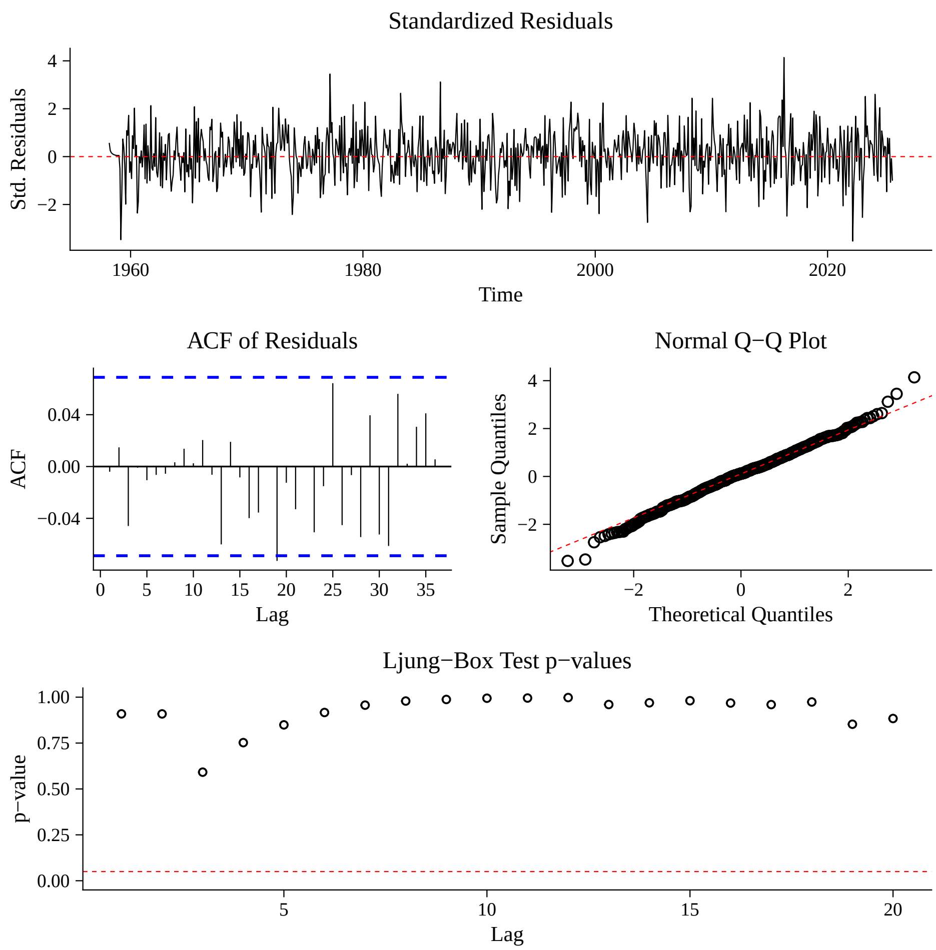


Figure 9: Model diagnostic for CO_2 data

6.5 Forecasting

We use SARIMA(1,1,1) $\times(0,1,1)_{12}$ model to forecast future monthly mean carbon dioxide values. Table 9 shows the estimated values for the future 12 months with the corresponding 95% confidence interval.

Table 9 and figure 10 show that the future monthly mean carbon dioxide value will still have a strong increasing trend with seasonality. It follows the same periodic fluctuation as previously, with the peak value in May, and the bottom value in September. In the year 2025-2026, the peak value was 432.59 in May and 424.31 in September. In general there is still an upward trend.

Month	Forecast CO2	Lower 95% CI	Upper 95% CI
Sep 2025	424.31	423.69	424.93
Oct 2025	424.58	423.85	425.32
Nov 2025	426.15	425.33	426.97
Dec 2025	427.61	426.72	428.49
Jan 2026	428.79	427.85	429.74
Feb 2026	429.69	428.68	430.69
Mar 2026	430.40	429.34	431.46
Apr 2026	431.90	430.79	433.02
May 2026	432.59	431.43	433.75
Jun 2026	432.06	430.85	433.27
Jul 2026	430.26	429.00	431.51
Aug 2026	428.12	426.82	429.42

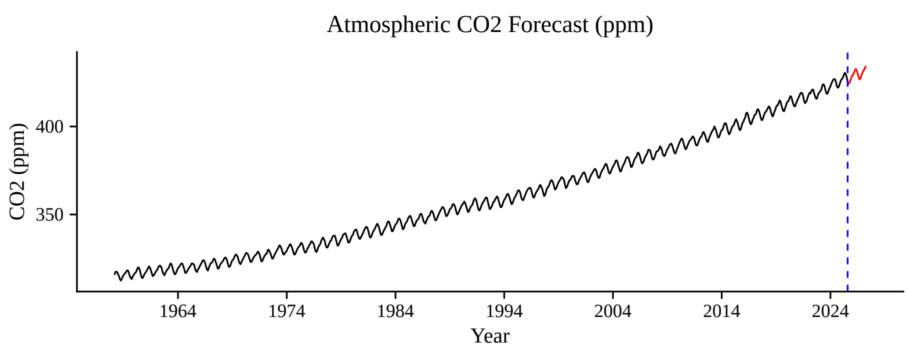


Figure 10: CO_2 Forecast

7 Discussion

In 2020, COVID-19 had a great impact on economics and led to a sharp GDP annual decline. The outlier of time series data points usually has a dominant influence on the forecast model, and further decrease the accuracy of forecasting, especially when the outlier is located at the end of the time series. The time series of the US data set we used in this project range from 1947.1.1 to 2025.1.1, the year 2020 is located relatively closed at the end of the time series. This gives the limitation of the ARIMA(2,0,2) model forecasting, the outlier and the location of the outlier might dominate the forecasting and lead to misleading results. One previous study suggested that the generalized ESD (Extreme Studentized Deviate) test is a capable way to catch the outlier, which can guide us to remove the certain outlier before we use the ARIMA model to do the forecasting [Rosner, 1983]. This might reduce or eliminate the deceptive forecasting of the ARIMA model when there are outliers in the time series dataset.

For the carbon dioxide dataset, we chose SARIMA(1,1,1) $\times(0,1,1)_{12}$ model to do the forecast, this result has been supported by some previous studies. There is one study that used the SARIMA(1,1,1) $\times(0,1,1)_{12}$ model on the differenced atmospheric Carbon Dioxide concentrations dataset, data is being measured at Bengaluru city in India, it suggests the same model as what we concluded in our study [Gogeri et al., 2023]. However, there are some limitations in our study. We assumed all the fluctuations were periodic before we fitted the SARIMA model, which might not always be true as some patterns are non-periodic. Also, the atmospheric carbon dioxide concentrations might have different seasonality when the locations differ, for example, the northern hemisphere has a strong periodic fluctuation compared with the southern hemisphere. This suggests that the SARIMA model choice is dependent on the locations [Gogeri et al., 2023]. Therefore, the model we concluded and the forecasting result might not be sufficient enough to draw any conclusions about the world's atmospheric Carbon Dioxide concentrations.

References

Mohamed R. Abonazel and Ahmed Ibrahim Abd-Elftah. Forecasting egyptian gdp using ARIMA models. *Reports on Economics and Finance*, 5(1):35–47, 2019.

Iranna Gogeri, K. C. Gouda, and T. Sumathy. Modeling and forecasting atmospheric carbon dioxide concentrations at bengaluru city in india. *Stochastic Environmental Research and Risk Assessment*, 37(12): 4685–4700, 2023.

Global Monitoring Laboratory. Mauna loa co2 monthly mean data, 2025. URL https://gml.noaa.gov/ccgg/trends/data.html.

X. B. Li and Xiaoxiao Zhang. A comparative study of statistical and machine learning models on carbon dioxide emissions prediction of china. *Environmental Science and Pollution Research*, 30(55):117485–117502, 2023.

M. Mamun Miah. Modelling and forecasting of GDP in bangladesh: An ARIMA approach. *Journal of Mechanics of Continua and Mathematical Sciences*, 14(3):245–256, 2019.

Bernard Rosner. Percentage points for a generalized esd many-outlier procedure. *Technometrics*, 25:165—172, 1983.

U.S. Bureau of Economic Analysis. Real gross domestic product, 2025. URL https://fred.stlouisfed.org/series/GDP.

Musundi Sammy Wabomba. Modeling and forecasting kenyan gdp using autoregressive integrated moving average (ARIMA) models. *Science Journal of Applied Mathematics and Statistics*, 4(2):64–71, 2016.